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UddeshayAnand

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Education

2024–2026 **M.Tech in Data Analytics**, *Dr. B. R. Ambedkar National Institute of Technology Jalandhar (NIT Jalandhar)*, *CGPA: 8.20/10*

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Statistical Computing, Databases, Data Structures & Algorithms

2020–2024 **B.E. in Mechanical Engineering**, *Panjab University*, Chandigarh, *CGPA: 8.18/10*

Relevant Coursework: Computer Programming, Statistics, Linear Algebra, Robotics, Mechatronics, Automatic Controls

Technical Skills

Languages	Python, SQL, C++, MATLAB
ML/DL	PyTorch, TensorFlow, Scikit-learn
AI/LLM	LangChain, RAG, Pinecone (Vector DB)
Data Tools	NumPy, Pandas, Matplotlib, Seaborn, Streamlit
Systems/Infra	Git, Linux, Database
Others	ROS/ROS2, Arduino, SolidWorks

Publications

Journal Articles (Under Review)

MBAN-AFBF: A Multi-Branch Gated Attention Network with Adaptive Spatial–Temporal–Spectral Feature Fusion for Motor Imagery EEG Decoding. **U. Anand**, K. K. Bharti, M. Sameer.

Discriminative Transformer Representation Learning from Complementary EEG Descriptors for Emotion Recognition. M. Sameer, K. K. Bharti, **U. Anand**.

Mobile Node Localization in Wireless Sensor Networks Using Graph Neural Networks and Deep Learning Models. A. Sharma, A. S. Shamkuwar, P. Bhardwaj, R. Kumar, N. Kumar, **U. Anand**, M. Singh.

Conference Papers (Accepted)

Multiband Attention Network for Motor Imagery EEG Classification (MBATNet). **U. Anand**, K. K. Bharti, M. Sameer. *MBCC 2026, IIT Mandi.*

ECOS-EEG: Evolutionary Channel Optimization for Subject-Independent EEG Emotion Recognition. M. Sameer, K. K. Bharti, **U. Anand**. *ICCEEE 2026, NIT Uttarakhand.*

Spatio-Temporal Node Localization in Wireless Sensor Networks Using Hybrid Recurrent-Graph Neural Network. A. Sharma, A. S. Shamkuwar, **U. Anand**, R. Kumar, N. Kumar, M. Singh. *ICSCCC, NIT Jalandhar.*

Experience

Mar 2026 – Present
Data Science Intern, *Oceaneering*
- Developed an end-to-end Content-Based Image Retrieval (CBIR) system using DINOv2 embeddings and FAISS vector indexing for semantic similarity search over a 100K-image dataset
- Built a full image processing pipeline (ingestion, preprocessing, feature extraction, indexing, retrieval) in Python and PyTorch, leveraging pretrained DINOv2 models via Hugging Face Transformers
- Developed an interactive Streamlit application enabling users to upload query images and view top-K retrieved results with similarity scores
- Evaluated retrieval quality using Top-K retrieval precision (Precision@5); applied Git-based version control for modular development

Jan 2024 – May 2024
Machine Learning Intern, *Artificial Limbs Manufacturing Corporation of India (ALIMCO)*
- Built a supervised regression pipeline to predict prosthetic limb position from EMG signals and real-time limb-position feedback, improving prosthetic control responsiveness
- Processed and analyzed biomedical sensor data using Python, NumPy, Pandas, and SciPy
- Trained and evaluated Linear Regression and Random Forest Regressor models with cross-validation; Linear Regression achieved the best performance with R^2 of approximately 0.98 (cross-validation performance above 0.96)
- Contributed to the design of an ML-assisted closed-loop prosthetic control concept combining EMG-based intent with motor-position feedback

Jul 2022 – Feb 2023
Data Analytics Intern, *Chandigarh Police (in collaboration with Infosys)*
- Analyzed cybercrime incident data (Jan–Jun 2022) to identify trends and anomalies, including a high prevalence of phishing attacks, supporting data-driven cybersecurity decision-making
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) using Python, Pandas, and NumPy
- Created visualizations using Matplotlib and Seaborn and generated analytical reports/dashboards to communicate findings to stakeholders

Academic Appointments

Aug 2025 – Nov 2025
Teaching Assistant – Machine Learning, *NIT Jalandhar*
Assisted in course delivery, assignment design, and evaluation for a graduate-level Machine Learning course

Jan 2025 – May 2025
Teaching Assistant – Python Programming, *NIT Jalandhar*
Assisted in course delivery and lab sessions for a graduate-level Python Programming course

Projects

PaperHelp – Hybrid Retrieval RAG System
- Built an end-to-end, production-style RAG application integrating LLMs with hybrid dense (FAISS) and sparse (BM25) retrieval via Reciprocal Rank Fusion
- Implemented cross-encoder reranking and confidence gating to reduce hallucinations through scoped refusal of low-confidence queries
- Engineered retrieval provenance and LLM-based grounding verification to audit chunk selection and validate answers against source documents
- Achieved 93% retrieval accuracy, 0.74 average answer similarity, and 82% grounding rate across 60 benchmark questions

Remaining Useful Life (RUL) Prediction for Aircraft Engines

- Processed the NASA C-MAPSS turbofan degradation dataset (100 engines, 21 sensor readings); performed EDA and feature reduction, removing low-variance, non-informative sensors
- Trained and compared Linear, Polynomial, Random Forest, XGBoost, and CatBoost regression models
- Random Forest achieved the best performance (RMSE: 29.56 cycles), demonstrating strong non-linear fault-behavior learning

Amazon ML Challenge 2025

- Built a multimodal (text + image) ML pipeline as part of a 5-member team to predict product prices across 75,000 products, evaluated using SMAPE
- Generated embeddings using Sentence Transformers; engineered KNN- and ANN-based features; trained a LightGBM Regressor for final prediction
- Achieved a SMAPE of 45%, ranking in the top 200 of 7,000+ participants nationwide

Achievements & Certifications

GATE 2024 Qualified – 91.73 percentile, Computer Science & Engineering

Amazon ML Challenge 2025 – Ranked in top 200 of 7,000+ participants nationwide